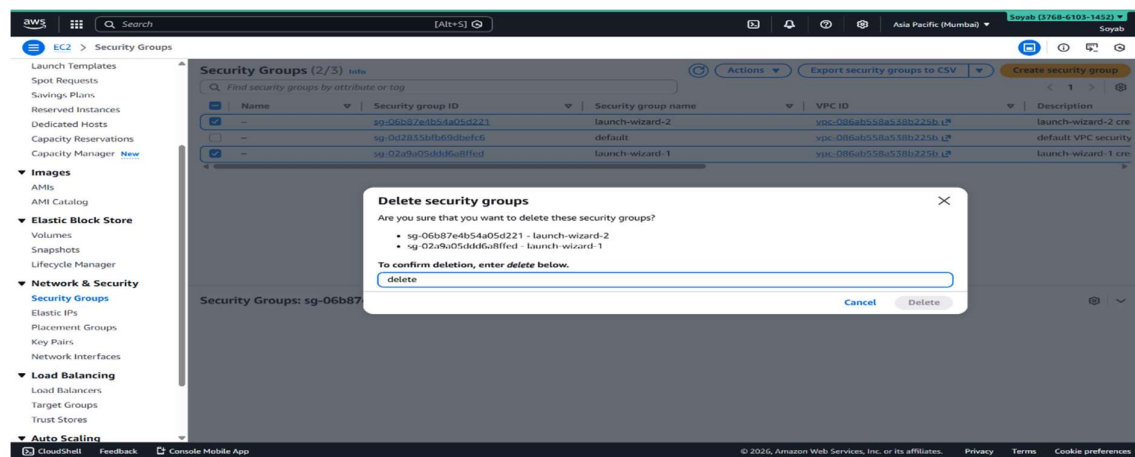
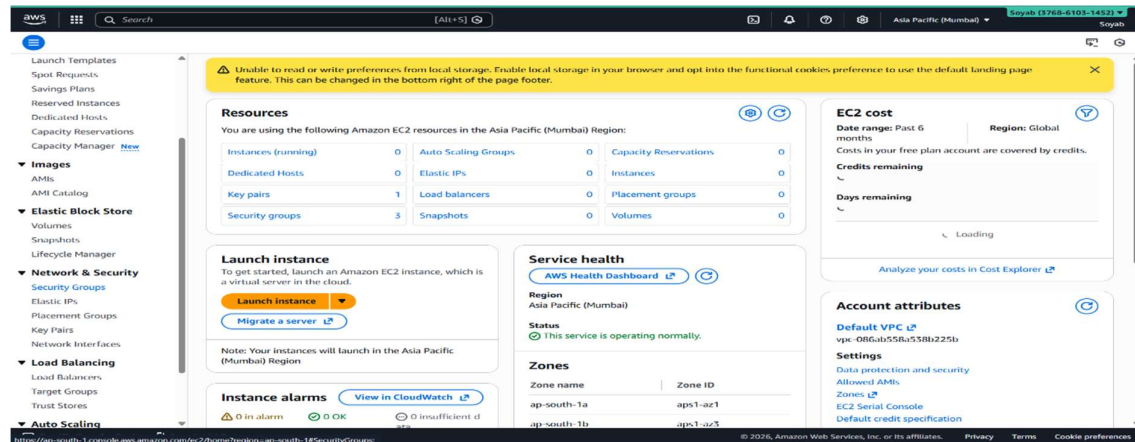


## ASIGNMENT NO :10

**PROBLEM STATEMENT :** Deploy a project from GitHub to EC2 by creating a new security

group and user data.

**STEP1 :-** First we go to EC2 Instance then we navigate to “Security Groups” under Network & Security.



**Step2:-** Then we create security group. In Create Security group we name it as we like then

give some description. then Inbound rules we add rule SSH,HTTP,HTTPS,Custom TCP, where for port range we keep it default for SSH,HTTP,HTTPS and for Custom TCP we

set it to 4000. We set source to “Anywhere-IPV4” 0.0.0.0/0

**Basic details**

Security group name info  
SoyabSecuritygroup

Description info  
Allows SSH access to developers

VPC info  
vpc-086ab558a538b225b

**Inbound rules**

| Type       | Protocol | Port range | Source   | Description - optional | Action |
|------------|----------|------------|----------|------------------------|--------|
| SSH        | TCP      | 22         | Anywh... | 0.0.0.0/0              | Delete |
| HTTP       | TCP      | 80         | Anywh... | 0.0.0.0/0              | Delete |
| HTTPS      | TCP      | 443        | Anywh... | 0.0.0.0/0              | Delete |
| Custom TCP | TCP      | 4000       | Anywh... | 0.0.0.0/0              | Delete |

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

**Step3 :- Then we Create Security group.**

**Outbound rules**

| Type        | Protocol | Port range | Destination | Description - optional | Action |
|-------------|----------|------------|-------------|------------------------|--------|
| All traffic | All      | All        | Custom      | 0.0.0.0/0              | Delete |

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

**Tags - optional**

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel Create security group

Security group (sg-01b66eabcc1495ac9 | SoyabSecuritygroup) was created successfully

**sg-01b66eabcc1495ac9 - SoyabSecuritygroup**

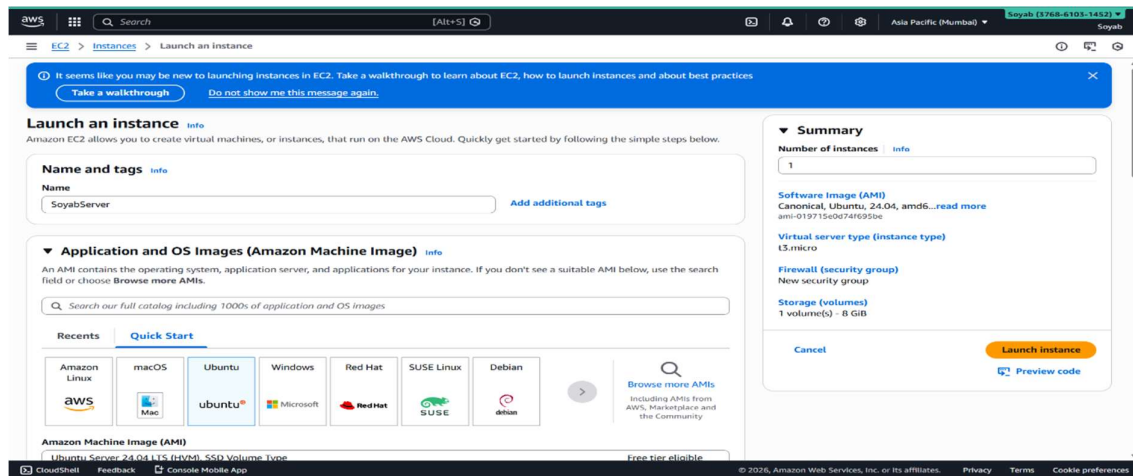
**Details**

|                                           |                                             |                                            |                                 |
|-------------------------------------------|---------------------------------------------|--------------------------------------------|---------------------------------|
| Security group name<br>SoyabSecuritygroup | Security group ID<br>sg-01b66eabcc1495ac9   | Description<br>security                    | VPC ID<br>vpc-086ab558a538b225b |
| Owner<br>376861031452                     | Inbound rules count<br>4 Permission entries | Outbound rules count<br>1 Permission entry |                                 |

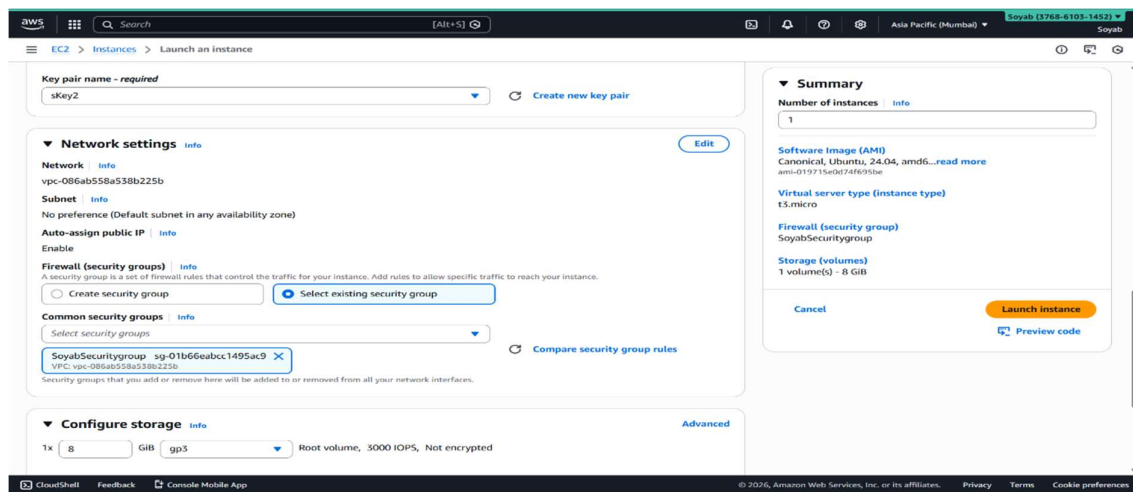
**Inbound rules (4)**

| Name | Security group rule ID | IP version | Type       | Protocol | Port range |
|------|------------------------|------------|------------|----------|------------|
| -    | sg-0176a205500a8fffd   | IPv4       | HTTP       | TCP      | 80         |
| -    | sg-0044ad46115c2c7ac   | IPv4       | SSH        | TCP      | 22         |
| -    | sg-0458979e64864ef4e   | IPv4       | Custom TCP | TCP      | 4000       |
| -    | sg-01e5d7ca2e7b69599   | IPv4       | HTTPS      | TCP      | 443        |

**Step4 :- Now we create a EC2 instance.**



**Step5 :- In Firewall(security group) we select our existing rule that we made earlier.**



**Step6 :- Now we scroll down to the advance details and then in user data we write the commands**

**” #!/bin/bash**

**apt-get update**

**apt-get upgrade**

**apt-get install-y nginx**

**systemctl start nginx**

**systemctl enable nginx**

**apt-get install-y git**

**curl-sL https://deb.nodesource.com/setup\_18.x | sudo-E bash**

apt-get install -y nodejs”

And launch the instance.

Metadata response hop limit: 2

Allow tags in metadata: Select

User data - optional: `#!/bin/bash  
apt-get update  
apt-get upgrade  
apt-get install -y nginx  
systemctl start nginx  
systemctl enable nginx  
apt-get install -y git  
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash  
apt-get install -y nodejs`

Summary:

- Number of instances: 1
- Software image (AMI): Canonical, Ubuntu, 24.04, amd64...read more
- Virtual server type (instance type): t3.micro
- Firewall (security group): SoyabSecuritygroup
- Storage (volumes): 1 volume(s) - 8 GiB

Buttons: Cancel, Launch instance, Preview code

Success: Successfully initiated launch of instance (i-0e46275e0d0ed1ae1)

Launch log

Next Steps: What would you like to do next with this instance, for example "create alarm" or "create backup"

- Create billing usage alerts
- Connect to your instance
- Connect an RDS database
- Create EBS snapshot policy
- Manage detailed monitoring
- Create Load Balancer
- Create AWS budget
- Manage CloudWatch alarms

step7 :- After creating EC2 Instances click on "CONNECT".

Connect to instance (i-0e46275e0d0ed1ae1)

Instance ID: i-0e46275e0d0ed1ae1 (SoyabServer)

VPC ID: vpc-086ab558a538b225b

Security groups: sg-01b66ebacc1495ac9 (SoyabSecuritygroup)

IAM role: -

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

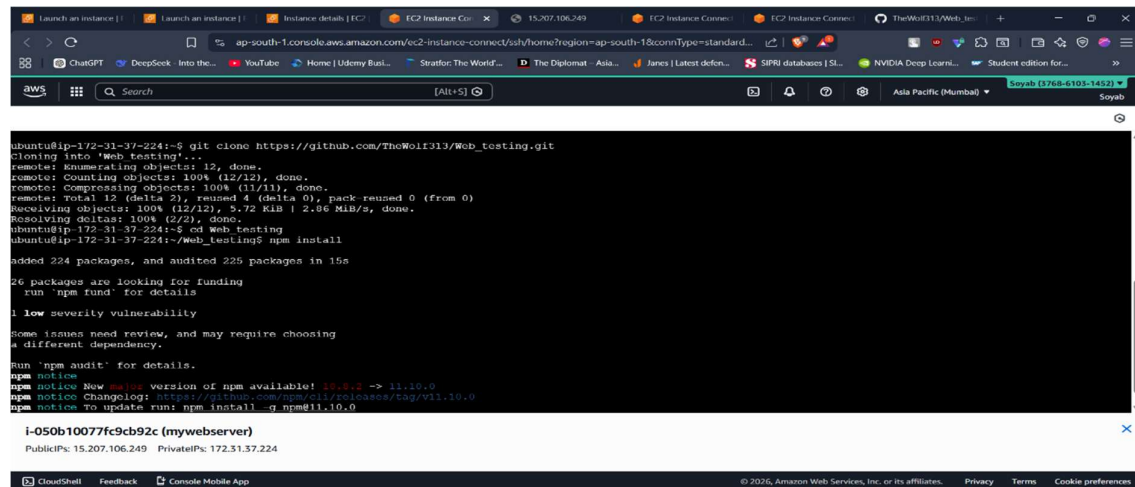
Connection type:

- Connect using a Public IP
- Connect using a Private IP

Public IPv4 address: 13.254.204.122

Username: ubuntu

**step8 :-** After connecting show new terminal.



```
ubuntu@ip-172-31-37-224:~$ git clone https://github.com/TheWolf313/Web_testing.git
Cloning into 'Web_testing'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 2), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 5.72 KiB | 2.56 MiB/s, done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-37-224:~$ cd Web_testing
ubuntu@ip-172-31-37-224:~/Web_testing$ npm install
added 224 packages, and audited 225 packages in 15s
26 packages are looking for funding
  run `npm fund` for details
1 low severity vulnerability
Some issues need review, and may require choosing
a different dependency.
Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.10.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.0
npm notice To update run: npm install -g npm@11.10.0

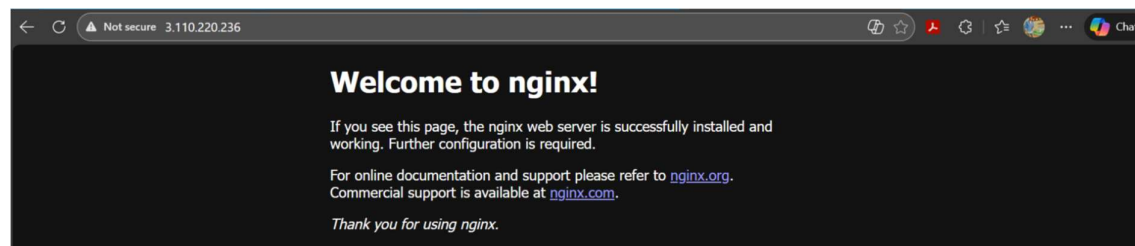
i-050b10077fc9cb92c (mywebserver)
PublicIPs: 15.207.106.249 PrivateIPs: 172.31.37.224
```

**STEP9 :-** RUN SOME COMMANDS "git clone <https://github.com/shawnitesh14-stack/demo1.git>"

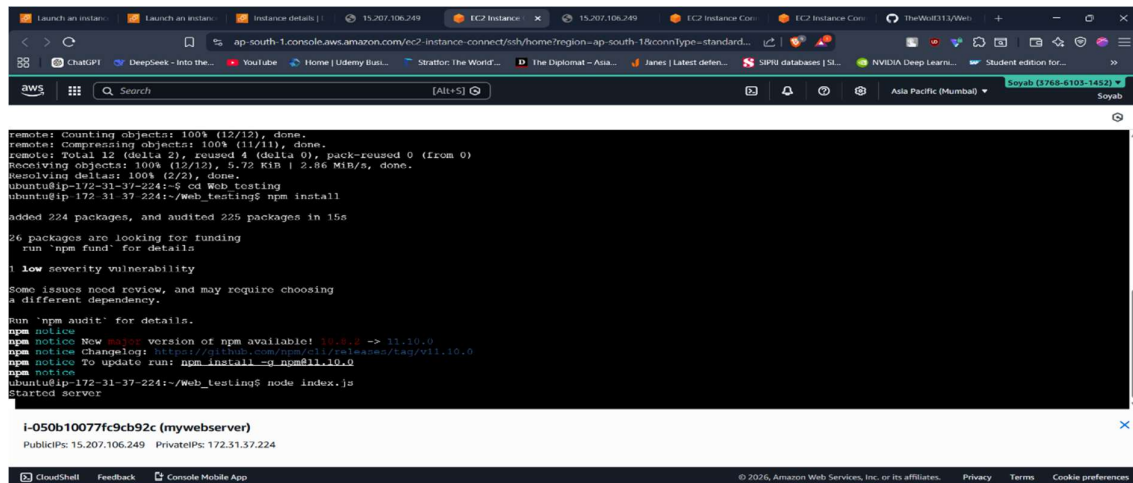


```
ubuntu@ip-172-31-42-42:~$ git clone https://github.com/shawnitesh14-stack/demo1.git
Cloning into 'demo1'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 9 (delta 3), reused 4 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (3/3), done.
ubuntu@ip-172-31-42-42:~$ ls
demo1
ubuntu@ip-172-31-42-42:~$ cd demo1
ubuntu@ip-172-31-42-42:~/demo1$ ls
index.html index.js next.html package.json previous.html
ubuntu@ip-172-31-42-42:~/demo1$
```

**Step10:-** Now we check if the server is running or not by opening incognito mode and going to the IP.



**step11 :-** after that we run some commands "npm install" and "node index.js" for start the server.



```
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 2), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 5.72 KiB | 2.86 MiB/s, done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-37-224:~$ cd /web_testing$ npm install
added 224 packages, and audited 225 packages in 15s
26 packages are looking for funding
  run 'npm fund' for details
1 low severity vulnerability
Some issues need review, and may require choosing
a different dependency.
run 'npm audit' for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.10.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.0
npm notice To update run: npm install -g npm@11.10.0
npm notice
ubuntu@ip-172-31-37-224:~$ cd /web_testing$ node index.js
Started server
```

i-050b10077fc9cb92c (mywebserver)  
PublicIPs: 15.207.106.249 PrivateIPs: 172.31.37.224

step12 :- Now we go to the IP 3.110.220.236:4000 to check our hosted project.



